孫子算經

The Mathematical Classic of Master Sun

anonymous, about the fourth or fifth century

Chinese text with modern punctuation, and a complete English translation

Chinese text: text as transcribed on Chinese Wikisource, with the Siku quanshu abstract, as transcribed on Chinese Wikisource (the pages and revisions used are listed at the end). Punctuation, division into units and English translation: AI-produced by Claude Opus 5.5 (Anthropic), 24–25 October 2026. The punctuated Chinese was checked by program against the Wikisource text: every character is reproduced, in order, and small print (commentary) stays small print. In the translation, commentary is given in square brackets with its author. The labels in the margin identify the units of the machine-readable file units.jsonl.

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The Mathematical Classic of Master Sun (Sunzi suanjing)

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四庫總目提要

Abstract from the General Catalogue of the Siku quanshu (Siku zongmu tiyao)

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案《隋書·經籍志》有《孫子算經》二卷，不注其名，亦不著其時代。《唐書·藝文志》稱李淳風注甄鸞《孫子算經》三卷。於孫子上冠以甄鸞，蓋如淳風之注《周髀算經》，因鸞所注更加辨論也。《隋書》論審度引《孫子算術》，蠶所吐絲為忽，十忽為秒，十秒為毫，十毫為氂，十氂為分，本書乃作十忽為一絲，十絲為一毫。又論嘉量引《孫子算術》，六粟為圭，十圭為秒，十秒為撮，十撮為勺，十勺為合。本書乃作十圭為一撮，十撮為一秒，十秒為一勺。考之夏侯陽《算經》引田曹、倉曹亦如本書，而《隋書》中所引與史傳往往多合。蓋古書傳本不一，校訂之儒各有據證，無妨參差互見也。

Note: The "Treatise on Classics and Books" (Jingji zhi) of the Sui shu lists a Sunzi suanjing in 2 juan, without noting its author's name and without recording its period. The "Treatise on Arts and Letters" (Yiwen zhi) of the Tang shu records Li Chunfeng's commentary on Zhen Luan's Sunzi suanjing, 3 juan. That Zhen Luan is put in front of Sunzi is presumably as with Chunfeng's commentary on the Zhoubi suanjing: it is because he added further discussion to what Luan had annotated. In discussing the regulation of length (shen du 審度), the Sui shu cites the Sunzi suanshu: the silk spat out by a silkworm is a hu (忽), 10 hu make a miao (秒), 10 miao make a hao (毫), 10 hao make a li (氂), 10 li make a fen (分); this book, however, has 10 hu make 1 si (絲), 10 si make 1 hao. Again, in discussing the standard capacity measure (jia liang 嘉量), it cites the Sunzi suanshu: 6 grains of millet (su 粟) make a gui (圭), 10 gui make a miao, 10 miao make a cuo (撮), 10 cuo make a shao (勺), 10 shao make a ge (合). This book, however, has 10 gui make 1 cuo, 10 cuo make 1 miao, 10 miao make 1 shao. Checking it against Xiahou Yang's Suanjing, its quotations from the Tiancao and the Cangcao are also as in this book, while what is quoted in the Sui shu often agrees with the histories and biographies. Presumably the transmitted copies of ancient books were not uniform, and the scholars who collated and corrected them each had their own evidence; there is no harm in the discrepancies appearing side by side.

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唐之選舉，算學孫子、五曹共限一歲習肄，於後來諸算術中特為近古，第不知孫子何許人。朱彝尊《曝書亭集·五曹算經跋》云，相傳其法出於孫武，然孫子別有《算經》，考古者存其説可爾。又有《孫子算經》跋云，首言度量所起，合乎兵法地生度，度生量，量生數之文。次言乘除之法設為之數，十三篇中所云廓地、分利、委積、遠輸、貴賣、兵役、分數比之《九章》方田、粟米、差分、商功、均輸、盈不足之目，往往相符，而要在得算多，多自然勝。以是知是編非僞託也云云。

In the Tang system of selection, in the School of Mathematics (suanxue 算學), the Sunzi and the Wucao together were allotted a limit of 1 year of study; among the mathematical works of later times it is especially close to antiquity. Only, it is not known who Sunzi was. Zhu Yizun's "Colophon to the Wucao suanjing" in the Pushuting ji says: "It is handed down that its methods derive from Sun Wu; but Sunzi has a separate Suanjing; those who investigate antiquity may preserve this account." He also has a colophon to the Sunzi suanjing, which says: "It first speaks of the origins of length and capacity measures (du liang 度量), which agrees with the passage of the Art of War: 'earth gives rise to measure (du), measure gives rise to quantity (liang), quantity gives rise to number (shu)'. Next it speaks of the methods of multiplication and division, setting up numbers for them. What the 13 chapters speak of—expanding territory, dividing the gains, stored provisions, long-distance transport, selling dear, military service, dividing and numbering—when compared with the headings of the Nine Chapters—Rectangular Fields, Millet and Rice, Distribution by Proportion, Assessing Works, Fair Transport, Excess and Deficit—often corresponds; and the essential

point lies in obtaining many calculations: with many, one naturally wins. By this one knows that this compilation is not a forgery," and so on.

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合二跋觀之，彝尊之意蓋以為确出於孫武。今考書內設問有云，長安洛陽相去九百里。又云，佛書二十九章，章六十三字，則後漢明帝以後人語。孫武春秋末人，安有是語乎？舊本久佚。今從《永樂大典》所載裒集編次，仍為三卷。其甄、李二家之注則不可復考，是則姚廣孝等割裂刊削之過矣。

Viewing the two colophons together, Yizun's opinion is presumably that it truly derives from Sun Wu. Now, examining the problems set within the book, one says: "Chang'an and Luoyang are 900 li (里) apart." Another says: "A Buddhist book of 29 chapters, each chapter of 63 characters." These, then, are the words of someone after Emperor Ming of the Later Han. Sun Wu was a man of the end of the Spring and Autumn period; how could he have such words? The old edition has long been lost. Now it has been collected and arranged in order from what is contained in the Yongle dadian, still in 3 juan. The commentaries of the two, Zhen and Li, can no longer be examined; this is the fault of Yao Guangxiao and others in cutting up and paring down.

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原序

Original Preface

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孫子曰：夫算者：天地之經緯，群生之元首，五常之本末，陰陽之父母，星辰之建號，三光之表裏，五行之準平，四時之終始，萬物之祖宗，六藝之綱紀。稽群倫之聚散，考二氣之降升，推寒暑之迭運，步遠近之殊同，觀天道精微之兆基，察地理從橫之長短，采神祇之所在，極成敗之符驗。窮道德之理，究性命之情。立規矩，準方圓，謹法度，約尺丈，立權衡，平重輕，剖毫釐，析泰絫。歷億載而不朽，施八極而無疆。散之者，富有餘；背之者，貧且窶。心開者，幼沖而即悟；意閉者，皓首而難精。夫欲學之者，必務量能揆己，志在所專，如是，則焉有不成者哉！

Master Sun said: As for calculation, it is the warp and weft of heaven and earth, the head of all living beings, the root and branches of the five constant virtues, the father and mother of yin and yang, the establisher of the names of the stars and constellations, the outside and inside of the three luminaries, the standard and level of the five phases, the end and beginning of the four seasons, the ancestor of the myriad things, and the guiding principle of the six arts. It examines the gathering and dispersing of all classes of beings, investigates the descent and ascent of the two qi, deduces the alternating revolutions of cold and heat, paces out the differences and samenesses of the far and the near, observes the fine and subtle first foundations of the Way of heaven, inspects the lengths, lengthwise and crosswise, of the patterns of the earth, seeks out where the gods of heaven and earth dwell, and fathoms the signs and proofs of success and failure. It exhausts the principles of the Way and virtue, and investigates the true conditions of human nature and destiny. It sets up the compass and the try-square and standardizes the square and the round; it makes careful the laws and measures and regulates the chi and the zhang; it sets up the weight and the beam and levels the heavy and the light; it cleaves the hao and the li, and divides the tai and the lei [Translator's note: 泰 tai is evidently an error for 黍 shu, the millet grain defined below as the origin of weights]. It passes through 100,000,000 years without decaying, and extends to the eight extremities without bounds. Those who spread it abroad are rich and have abundance; those who turn their backs on it are poor and destitute. Those whose minds are open understand it at once even in tender youth; those whose thoughts are closed find it hard to master even when white-haired. Whoever wishes to study it must strive to measure his abilities and assess himself, and set his will on what he devotes himself to; if so, how could there be anyone who does not succeed!

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卷上

Upper Juan

sunzi-001.9

度之所起，起于忽。欲知其忽，蠶吐絲為忽，十忽為一絲，十絲為一毫，十毫為一氂，十氂為一分，十分為一寸，十寸為一尺，十尺為一丈，十丈為一引，五十引為一端，四十尺為一匹，六尺為一步，二百四十步為一畝，三百步為一里。

The origin of length measure (du 度): it arises from the hu (忽). If you wish to know the hu: the silk spat out by a silkworm is a hu; 10 hu make 1 si (絲); 10 si make 1 hao (毫); 10 hao make 1 li (氂); 10 li make 1 fen (分); 10 fen make 1 cun (寸); 10 cun make 1 chi (尺); 10 chi make 1 zhang (丈); 10 zhang make 1 yin (引); 50 yin make 1 duan (端); 40 chi make 1 pi (匹); 6 chi make 1 bu (步); 240 bu make 1 mu (畝); 300 bu make 1 li (里).

sunzi-001.10

稱之所起，起于黍。十黍為一絫，十絫為一銖，二十四銖為一兩，十六兩為一觔，三十觔為一鈞，四鈞為一石。

The origin of weighing (cheng 稱): it arises from the millet grain (shu 黍). 10 millet grains make 1 lei (絫); 10 lei make 1 zhu (銖); 24 zhu make 1 liang (兩); 16 liang make 1 jin (觔); 30 jin make 1 jun (鈞); 4 jun make 1 dan (石).

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量之所起，起于粟。六粟為一圭，十圭為一撮，十撮為一抄，十抄為一勺，十勺為一合，十合為一升，十升為一斗，十斗為一斛，十斛得六千萬粟。所以得知者，六粟為一圭，十圭六十粟為一撮，十撮六百粟為一抄，十抄六千粟為一勺，十勺六萬粟為一合，十合六十萬粟為一升，十升六百萬粟為一斗，十斗六千萬粟為一斛，十斛六億粟，百斛六兆粟，千斛六京粟，萬斛六垓粟，十萬斛六秭粟，百萬斛六穰粟，千萬斛六溝粟，萬萬斛為一億，六澗粟，十億斛六正粟，百億斛六載粟。

The origin of capacity measure (liang 量): it arises from the grain of millet (su 粟). 6 grains make 1 gui (圭); 10 gui make 1 cuo (撮); 10 cuo make 1 chao (抄); 10 chao make 1 shao (勺); 10 shao make 1 ge (合); 10 ge make 1 sheng (升); 10 sheng make 1 dou (斗); 10 dou make 1 hu (斛); 10 hu yield 60,000,000 grains. The way this is known: 6 grains make 1 gui; 10 gui, 60 grains, make 1 cuo; 10 cuo, 600 grains, make 1 chao; 10 chao, 6,000 grains, make 1 shao; 10 shao, 60,000 grains, make 1 ge; 10 ge, 600,000 grains, make 1 sheng; 10 sheng, 6,000,000 grains, make 1 dou; 10 dou, 60,000,000 grains, make 1 hu; 10 hu, 6 yi (億) grains; 100 hu, 6 zhao (兆) grains; 1,000 hu, 6 jing (京) grains; 10,000 hu, 6 gai (垓) grains; 100,000 hu, 6 zi (秭) grains; 1,000,000 hu, 6 rang (穰) grains; 10,000,000 hu, 6 gou (溝) grains; 10,000 times 10,000 hu make 1 yi: 6 jian (澗) grains; 10 yi hu, 6 zheng (正) grains; 100 yi hu, 6 zai (載) grains.

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凡大數之法：萬萬曰億，萬萬億曰兆，萬萬兆曰京，萬萬京曰垓，萬萬垓曰秭，萬萬秭曰穰，萬萬穰曰溝，萬萬溝曰澗，萬萬澗曰正，萬萬正曰載。

In general, the rule for large numbers: 10,000 times 10,000 is called yi (億); 10,000 times 10,000 yi is called zhao (兆); 10,000 times 10,000 zhao is called jing (京); 10,000 times 10,000 jing is called gai (垓); 10,000 times 10,000 gai is called zi (秭); 10,000 times 10,000 zi is called rang (穰); 10,000 times 10,000 rang is called gou (溝); 10,000 times 10,000 gou is called jian (澗); 10,000 times 10,000 jian is called zheng (正); 10,000 times 10,000 zheng is called zai (載).

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周三，徑一，方五，邪七。見邪求方，五之，七而一；見方求邪，七之，五而一。

Circumference 3, diameter 1; square (fang 方) 5, diagonal (xie 邪) 7. Given the diagonal, to find the square: multiply it by 5 and divide by 7; given the square, to find the diagonal: multiply it by 7 and divide by 5.

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白銀方寸重一十四兩。玉方寸重一十兩。銅方寸重七兩半。鉛方寸重九兩半。鐵方寸重七兩。石方寸重三兩。

A cubic cun (fang cun 方寸) of white silver weighs 14 liang. A cubic cun of jade weighs 10 liang. A cubic cun of copper weighs 7 1/2 liang. A cubic cun of lead weighs 9 1/2 liang. A cubic cun of iron weighs 7 liang. A cubic cun of stone weighs 3 liang.

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凡算之法：先識其位，一從十橫，百立千僵，千十相望，萬百相當。（案：萬百原本訛作百萬，今據《夏侯陽算經》改正。）

The rule of calculation in general: first know the places (wei 位). Units are vertical, tens horizontal; hundreds stand, thousands lie down; thousands and tens look at each other, ten-thousands and hundreds correspond. (Siku editors: "ten-thousands and hundreds" (萬百) was miswritten as 百萬 in the original edition; now corrected on the basis of the Xiahou Yang suanjing.)

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凡乘之法：重置其位，上下相觀，頭位有十步至十，有百步至百，有千步至千，以上命下，所得之數列于中。言十即過，不滿，自如頭位。乘訖者，先去之；下位乘訖者，則俱退之。六不積，五不隻。上下相乘，至盡則已。

The rule of multiplication in general: set their places in two layers (chong zhi 重置), the upper and the lower looking at each other. If the leading place (tou wei 頭位) has tens, step (bu 步) to the tens; if it has hundreds, step to the hundreds; if it has thousands, step to the thousands. With the upper, name (ming 命) the lower; the number obtained is laid out in the middle. If it speaks of tens, it passes on; if it is not full, it remains as at the leading place. When the multiplying is finished, first take it away; when the lower place has finished multiplying, then move them all back (tui 退). 6 is not piled up; 5 is not single. Multiply upper and lower by each other, and stop when they are exhausted.

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凡除之法：與乘正異。乘得在中央，除得在上方。假令六為法，百為實，以六除百，當進之二等，令在正百下。以六除一，則法多而實少，不可除，故當退就十位。以法除實，言一六而折百為四十，故可除。若實多法少，自當百之，不當復退，故或步法十者，置于十百位（頭位有空絕者，法退二位。）餘法皆如乘時。實有餘者，以法命之，以法為母，實餘為子。

The rule of division in general: it is exactly the opposite of multiplication. What multiplication obtains is in the middle; what division obtains is in the upper position. Suppose 6 is the divisor (fa) and 100 the dividend (shi). Dividing 100 by 6, one should advance (jin 進) it 2 ranks (deng 等), so that it is directly under the hundred. Dividing 1 by 6, the divisor is more and the dividend less, so it cannot be divided; therefore one should move it back to the tens place. Dividing the dividend by the divisor, one says "1 times 6" and reduces the 100 to 40; therefore it can be divided. If the dividend is more and the divisor less, it naturally belongs at the hundreds and should not be moved back again. Therefore, where one steps the divisor by tens, it is set at the ten-hundred place [Translator's note: the text appears defective here] (If the leading place has an empty gap, the divisor is moved back 2 places.) The rest of the procedure is all as in multiplication. If the dividend has a remainder, name it by the divisor: take the divisor as the denominator (mu 母) and the remainder of the dividend as the numerator (zi 子).

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以粟求糲米，三之，五而一。以糲米求粟，五之，三而一。以糲米求飯，五之，二而一。以粟米求糲飯，六之，四而一。以糲飯求糲米，二之，五而一。以□米求飯，八之，四而一。

From millet (su 粟), to find coarse hulled grain (li mi 糲米): multiply it by 3 and divide by 5. From coarse hulled grain, to find millet: multiply it by 5 and divide by 3. From coarse hulled grain, to find cooked grain (fan 飯): multiply it by 5 and divide by 2. From millet grain (su mi 粟米), to find cooked coarse grain (li fan 糲飯): multiply it by 6 and divide by 4. From cooked coarse grain, to find coarse hulled grain: multiply it by 2 and divide by 5. From [lost character] hulled grain (□米), to find cooked grain: multiply it by 8 and divide by 4.

sunzi-001.19

十分減一者，以二乘，二十除；減二者，以四乘，二十除；減三者，以六乘，二十除；減四者，以八乘，二十除；減五者，以十乘，二十除；減六者，以十二乘，二十除；減七者，以十四乘，二十除；減八者，以十六乘，二十除；減九者，以十八乘，二十除。九分減一者，以二乘，十八除。八分減一者，以二乘，十六除。七分減一者，以二乘，十四除。六分減一者，以二乘，十二除。五分減一者，以二乘，十除。

Where 1 part in 10 is subtracted, multiply by 2 and divide by 20; where 2 are subtracted, multiply by 4 and divide by 20; where 3 are subtracted, multiply by 6 and divide by 20; where 4 are subtracted, multiply by 8 and divide by 20; where 5 are subtracted, multiply by 10 and divide by 20; where 6 are subtracted, multiply by 12 and divide by 20; where 7 are subtracted, multiply by 14 and divide by 20; where 8 are subtracted, multiply by 16 and divide by 20; where 9 are subtracted, multiply by 18 and divide by 20. Where 1 part in 9 is subtracted, multiply by 2 and divide by 18. Where 1 part in 8 is subtracted, multiply by 2 and divide by 16. Where 1 part in 7 is subtracted, multiply by 2 and divide by 14. Where 1 part in 6 is subtracted, multiply by 2 and divide by 12. Where 1 part in 5 is subtracted, multiply by 2 and divide by 10.

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九九八十一，自相乘得幾何？答曰：六千五百六十一。

9 times 9 is 81; multiplied by itself, how much does it give? Answer: 6,561.

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術曰：重置其位，以上八呼下八，八八六十四，即下六千四百于中位；以上八呼下一，一八如八，即于中位下八十，退下位一等，收上頭位八十（案：原本脫「上」字，今補。）；以上位一（案：上位原本訛作「頭位」，今改正。）呼下八，一八如八，即于中位下八十；以上一呼下一，一一如一，即于中位下一，上下位俱收，中位即得六千五百六十一。

Method: Set their places in two layers. With the upper 8 call (hu 呼) the lower 8: 8 times 8 is 64; then set down 6,400 in the middle place (zhong wei 中位). With the upper 8 call the lower 1: 1 times 8 is 8; then set down 80 in the middle place. Move the lower place back 1 rank; remove (shou 收) the 80 of the upper leading place (Siku editors: the original edition lacks the character 上”upper”; now supplied.); with the 1 of the upper place (Siku editors: ”upper place” (上位) was miswritten as ”leading place” (頭位) in the original edition; now corrected.) call the lower 8: 1 times 8 is 8; then set down 80 in the middle place. With the upper 1 call the lower 1: 1 times 1 is 1; then set down 1 in the middle place. Remove both the upper and lower places; the middle place then gives 6,561.

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六千五百六十一，九人分之。問：人得幾何？答曰：七百二十九。

6,561 is divided among 9 persons. Question: how much does each person get? Answer: 729.

sunzi-001.23

術曰：先置六千五百六十一于中位，為實，下列九人為法，頭位置七百（案：原本脫上字，今補。），以上七呼下九，七九六十三，即除中位六千三百，退下位一等，即上位置二十（案：上位原本訛作頭位，今改正。），以上二呼下九，二九一十八，即除中位一百八十，又更退下位一等，即上位更置九（案：上位原本亦訛作頭位，今改正。），即以上九呼下九，九九八十一，即除中位八十一，中位並盡，收下位，頭位所得即人之所得，自八八六十四至一一如一，並準此。

Method: First set 6,561 in the middle place, as the dividend (shi); below, lay out the 9 persons as the divisor (fa). In the leading place set 700 (Siku editors: the original edition lacks the character 上”upper”; now supplied.). With the upper 7 call the lower 9: 7 times 9 is 63; then subtract 6,300 from the middle place. Move the lower place back 1 rank; then in the upper place set 20 (Siku editors: ”upper place” (上位) was miswritten as ”leading place” (頭位) in the original edition; now corrected.). With the upper 2 call the lower 9: 2 times 9 is 18; then subtract 180 from the middle place. Again move the lower place back a further 1 rank; then in the upper place set a further 9 (Siku editors: ”upper place” (上位) was also miswritten as ”leading place” (頭位) in the original edition; now corrected.). Then with the upper 9 call the lower 9: 9 times 9 is 81; then subtract 81 from the middle place. The middle place is entirely exhausted; remove the lower place. What is obtained in the leading place is what each person gets. From ”8 times 8 is 64” to ”1 times 1 is 1”, all follow this.

sunzi-001.24

八九七十二，自相乘，得五千一百八十四，八人分之，人得六百四十八。七九六十三，自相乘，得三千九百六十九，七人分之，人得五百六十七。六九五十四，自相乘，得二千九百一十六，六人分之，人得四百八十六。五九四十五，自相乘，得二千二十五，五人分

之，人得四百五。四九三十六，自相乘，得一千二百九十六，四人分之，人得三百二十四。三九二十七，自相乘，得七百二十九，三人分之，人得二百四十三。二九一十八，自相乘，得三百二十四，二人分之，人得一百六十二。一九如九，自相乘，得八十一，一人得八十一。

8 times 9 is 72; multiplied by itself, it gives 5,184; divided among 8 persons, each person gets 648. 7 times 9 is 63; multiplied by itself, it gives 3,969; divided among 7 persons, each person gets 567. 6 times 9 is 54; multiplied by itself, it gives 2,916; divided among 6 persons, each person gets 486. 5 times 9 is 45; multiplied by itself, it gives 2,025; divided among 5 persons, each person gets 405. 4 times 9 is 36; multiplied by itself, it gives 1,296; divided among 4 persons, each person gets 324. 3 times 9 is 27; multiplied by itself, it gives 729; divided among 3 persons, each person gets 243. 2 times 9 is 18; multiplied by itself, it gives 324; divided among 2 persons, each person gets 162. 1 times 9 is 9; multiplied by itself, it gives 81; 1 person gets 81.

sunzi-001.25

右九九一條，得四百五，自相乘，得一十六萬四千二十五，九人分之，人得八千二百二十五。

The section of nines above, as one item, gives 405; multiplied by itself, it gives 164,025; divided among 9 persons, each person gets 8,225.

sunzi-001.26

八八六十四，自相乘，得四千九十六，八人分之，人得五百一十二。七八五十六，自相乘，得三千一百三十六，七人分之，人得四百四十八。六八四十八，自相乘，得二千三百四，六人分之，人得三百八十四。五八四十，自相乘，得一千六百，五人分之，人得三百二十。四八三十二，自相乘，得一千二十四，四人分之，人得二百五十六。三八二十四，自相乘，得五百七十六，三人分之，人得一百九十二。二八十六，自相乘，得二百五十六，二人分之，人得一百二十八。一八如八，自相乘，得六十四，一人得六十四。

8 times 8 is 64; multiplied by itself, it gives 4,096; divided among 8 persons, each person gets 512. 7 times 8 is 56; multiplied by itself, it gives 3,136; divided among 7 persons, each person gets 448. 6 times 8 is 48; multiplied by itself, it gives 2,304; divided among 6 persons, each person gets 384. 5 times 8 is 40; multiplied by itself, it gives 1,600; divided among 5 persons, each person gets 320. 4 times 8 is 32; multiplied by itself, it gives 1,024; divided among 4 persons, each person gets 256. 3 times 8 is 24; multiplied by itself, it gives 576; divided among 3 persons, each person gets 192. 2 times 8 is 16; multiplied by itself, it gives 256; divided among 2 persons, each person gets 128. 1 times 8 is 8; multiplied by itself, it gives 64; 1 person gets 64.

sunzi-002.1

右八八一條，得二百八十八，自相乘，得八萬二千九百四十四，八人分之，人得一萬三百六十八。

The above one series of ”eight eights” gives 288; multiplied by itself, it gives 82,944; divided among 8 persons, each person gets 10,368.

sunzi-002.2

七七四十九，自相乘，得二千四百一，七人分之，人得三百四十三。六七四十二，自相乘，得一千七百六十四，六人分之，人得二百九十四。五七三十五，自相乘，得一千二百二十五，五人分之，人得二百四十五。四七二十八，自相乘，得七百八十四，四人分之，人得一百九十六。三七二十一，自相乘，得四百四十一，三人分之，人得一百四十七。二七一十四，自相乘，得一百九十六，二人分之，人得九十八。一七如七，自相乘，得四十九，一人得四十九。右七七一條，得一百九十六，自相乘，得三萬八千四百一十六，七人分之，人得五千四百八十八。

Seven sevens are 49; multiplied by itself, it gives 2401; divided among 7 persons, each person gets 343. Six sevens are 42; multiplied by itself, it gives 1764; divided among 6 persons, each person gets 294. Five sevens are 35; multiplied by itself, it gives 1225; divided among 5 persons, each person gets 245. Four sevens are 28; multiplied by itself, it gives 784; divided among 4 persons,

each person gets 196. Three sevens are 21; multiplied by itself, it gives 441; divided among 3 persons, each person gets 147. Two sevens are 14; multiplied by itself, it gives 196; divided among 2 persons, each person gets 98. One seven makes 7; multiplied by itself, it gives 49; one person gets 49. The above one series of ”seven sevens” gives 196; multiplied by itself, it gives 38,416; divided among 7 persons, each person gets 5488.

sunzi-002.3

六六三十六，自相乘，得一千二百九十六，六人分之，人得二百一十六。五六三十，自相乘，得九百，五人分之，人得一百八十。四六二十四，自相乘，得五百七十六，四人分之，人得一百四十四。三六一十八，自相乘，得三百二十四，三人分之，人得一百八。二六一十二，自相乘，得一百四十四，二人分之，人得七十二。一六如六，自相乘，得三十六，一人得三十六。右六六一條，得一百二十六，自相乘，得一萬五千八百七十六，六人分之，人得二千六百四十六。

Six sixes are 36; multiplied by itself, it gives 1296; divided among 6 persons, each person gets 216. Five sixes are 30; multiplied by itself, it gives 900; divided among 5 persons, each person gets 180. Four sixes are 24; multiplied by itself, it gives 576; divided among 4 persons, each person gets 144. Three sixes are 18; multiplied by itself, it gives 324; divided among 3 persons, each person gets 108. Two sixes are 12; multiplied by itself, it gives 144; divided among 2 persons, each person gets 72. One six makes 6; multiplied by itself, it gives 36; one person gets 36. The above one series of ”six sixes” gives 126; multiplied by itself, it gives 15,876; divided among 6 persons, each person gets 2646.

sunzi-002.4

五五二十五，自相乘，得六百二十五，五人分之，人得一百二十五。四五二十，自相乘，得四百，四人分之，人得一百。三五一十五，自相乘，得二百二十五，三人分之，人得七十五。二五一十，自相乘，得一百，二人分之，得五十。一五如五，自相乘，得二十五，一人得二十五。右五五一條，得七十五，自相乘，得五千六百二十五，五人分之，人得一千一百二十五。

Five fives are 25; multiplied by itself, it gives 625; divided among 5 persons, each person gets 125. Four fives are 20; multiplied by itself, it gives 400; divided among 4 persons, each person gets 100. Three fives are 15; multiplied by itself, it gives 225; divided among 3 persons, each person gets 75. Two fives are 10; multiplied by itself, it gives 100; divided among 2 persons, it gives 50. One five makes 5; multiplied by itself, it gives 25; one person gets 25. The above one series of ”five fives” gives 75; multiplied by itself, it gives 5625; divided among 5 persons, each person gets 1125.

sunzi-002.5

四四一十六，自相乘，得二百五十六，四人分之，人得六十四。三四一十二，自相乘，得一百四十四，三人分之，人得四十八。二四如八，自相乘，得六十四，二人分之，人得三十二。一四如四，自相乘，得一十六，一人得一十六。右四四一條，得四十，自相乘，得一千六百，四人分之，人得四百。

Four fours are 16; multiplied by itself, it gives 256; divided among 4 persons, each person gets 64. Three fours are 12; multiplied by itself, it gives 144; divided among 3 persons, each person gets 48. Two fours make 8; multiplied by itself, it gives 64; divided among 2 persons, each person gets 32. One four makes 4; multiplied by itself, it gives 16; one person gets 16. The above one series of ”four fours” gives 40; multiplied by itself, it gives 1600; divided among 4 persons, each person gets 400.

sunzi-002.6

三三如九，自相乘，得八十一，三人分之，人得二十七。二三如六，自相乘，得三十六，二人分之，人得一十八。一三如三，自相乘，得九，一人得九。右三三一條，得一十八，自相乘，得三百二十四，三人分之，人得一百八。

Three threes make 9; multiplied by itself, it gives 81; divided among 3 persons, each person gets 27. Two threes make 6; multiplied by itself, it gives 36; divided among 2 persons, each person gets 18. One three makes 3; multiplied by itself, it gives 9; one person gets 9. The above one series of

”three threes” gives 18; multiplied by itself, it gives 324; divided among 3 persons, each person gets 108.

sunzi-002.7

二二如四，自相乘，得一十六，二人分之，人得八。一二如二，自相乘，得四，一人得四。右二二一條，得六，自相乘，得三十六，二人分之，人得一十八。

Two twos make 4; multiplied by itself, it gives 16; divided among 2 persons, each person gets 8. One two makes 2; multiplied by itself, it gives 4; one person gets 4. The above one series of ”two twos” gives 6; multiplied by itself, it gives 36; divided among 2 persons, each person gets 18.

sunzi-002.8

一一如一，自相乘，得一，一乘不長。右從九九至一一，總成一千一百五十五，自相乘，得一百三十三萬四千二十五，九人分之，人得一十四萬八千二百二十五。

One one makes 1; multiplied by itself, it gives 1; multiplying by one produces no increase. The above, from ”nine nines” to ”one one”, altogether makes 1155; multiplied by itself, it gives 1,334,025; divided among 9 persons, each person gets 148,225.

sunzi-002.9

以九乘一十二，得一百八，六人分之，人得一十八。以二十七乘三十六，得九百七十二，一十八人分之，人得五十四。以八十一乘一百八，得八千七百四十八，五十四人分之，人得六十二。以二百四十三乘三百二十四，得七萬八千七百三十二，一百六十二人分之，人得四百八十六。以七百二十九乘九百七十二，得七十萬八千五百八十八，四百八十六人分之，人得一千四百五十八。以二千一百八十七乘二千九百一十六，得六百三十七萬七千二百九十二，一千四百五十八人分之，得四千三百七十四。以六千五百六十一乘八千七百四十八，得五千七百三十九萬五千六百二十八，四千三百七十四人分之，人得一萬三千一百二十二。

Using 9 to multiply 12 gives 108; divided among 6 persons, each person gets 18. Using 27 to multiply 36 gives 972; divided among 18 persons, each person gets 54. Using 81 to multiply 108 gives 8748; divided among 54 persons, each person gets 62 [Translator’s note: 8748 ÷ 54 = 162]. Using 243 to multiply 324 gives 78,732; divided among 162 persons, each person gets 486. Using 729 to multiply 972 gives 708,588; divided among 486 persons, each person gets 1458. Using 2187 to multiply 2916 gives 6,377,292; divided among 1458 persons, this gives 4374. Using 6561 to multiply 8748 gives 57,395,628; divided among 4374 persons, each person gets 13,122.

sunzi-002.10

以一萬九千六百八十三乘二萬六千二百四十四，得五億一千六百五十六萬六百五十二，一萬三千一百二十二人分之，人得三萬九千三百六十六。以五萬九千四十九乘七萬八千七百三十二，得四十六億四千九百四萬五千八百六十八，三萬九千三百六十六人分之，人得一十一萬八千九十八。以一十七萬七千一百四十七乘二十三萬六千一百九十六，得四百一十八億四千一百四十一萬二千八百一十二，一十一萬八千九十八人分之，得三十五萬四千二百九十四。以五十三萬一千四百四十一乘七十萬八千五百八十八，得三千七百六十五億七千二百七十一萬五千三百八，三十五萬四千二百九十四人分之，人得一百六萬二千八百八十二。

Using 19,683 to multiply 26,244 gives 516,560,652; divided among 13,122 persons, each person gets 39,366. Using 59,049 to multiply 78,732 gives 4,649,045,868; divided among 39,366 persons, each person gets 118,098. Using 177,147 to multiply 236,196 gives 41,841,412,812; divided among 118,098 persons, this gives 354,294. Using 531,441 to multiply 708,588 gives 376,572,715,308; divided among 354,294 persons, each person gets 1,062,882.

sunzi-002.11

卷中

Middle Volume (juan zhong)

sunzi-002.12

今有一十八分之一十二。問：約之得幾何？答曰：三分之二。

Suppose there is 12/18. Question: on reducing (yue) it, how much does one get? Answer: 2/3.

sunzi-002.13

術曰：置十八分在下，一十二分在上，副置二位，以少減多，等數得六為法，約之即得。

Method: Place the 18 parts below and the 12 parts above; set out the two positions in duplicate (fuzhi), and subtract the lesser from the greater; the equal number (dengshu) obtained is 6, which is taken as the divisor (fa); reduce by it, and one obtains [the answer] at once.

sunzi-003.1

今有三分之一、五分之二。問：合之二得幾何？答曰：一十五分之十一。

Suppose there is 1/3 and 2/5. Question: combining the two, how much is obtained? Answer: 11/15.

sunzi-003.2

術曰：置三分五分在右方，之一之二在左方，母互乘子，五分之二得六，三分之一得五，并之，得一十一為實；又方二母相乘，得一十五為法。不滿法，以法命之，即得。

Method: Place '3 parts' and '5 parts' on the right side, and '1 of them' and '2 of them' on the left side. The denominators (mu) cross-multiply (hucheng) the numerators (zi): 2/5 gives 6 and 1/3 gives 5. Combine them, obtaining 11 as the dividend (shi). Further, multiply together the two denominators of the side, obtaining 15 as the divisor (fa). It does not fill the divisor: name it by the divisor, and [the answer] is obtained.

sunzi-003.3

今有九分之八，減其五分之一。問：餘幾何？答曰：四十五分之三十一。

Suppose there is 8/9, and 1/5 is subtracted from it. Question: how much remains? Answer: 31/45.

sunzi-003.4

術曰：置九分五分在右方，之八之一在左方，母互乘子，五分之一得九，九分之八得四十，以少減多，餘三十一，為實；母相乘，得四十五，為法。不滿法，以法命之，即得。

Method: Place '9 parts' and '5 parts' on the right side, and '8 of them' and '1 of them' on the left side. The denominators cross-multiply the numerators: 1/5 gives 9 and 8/9 gives 40. Subtract the lesser from the greater, leaving 31, as the dividend. Multiply the denominators together, obtaining 45, as the divisor. It does not fill the divisor: name it by the divisor, and [the answer] is obtained.

sunzi-003.5

今有三分之一，三分之二，四分之三。問：減多益少，幾何而平？答曰：減四分之三者二，減三分之二者一，并以益三分之一，而各平于一十二分之七。

Suppose there is 1/3, 2/3, 3/4. Question: subtracting from the greater and augmenting the lesser, by how much are they made level? Answer: Subtract 2 from the 3/4, subtract 1 from the 2/3, combine [these] and augment the 1/3 with them, and each is level at 7/12.

sunzi-003.6

術曰：置三分三分四分在右方，之一之二之三在左方，母互乘子，副并得六十三。置右為平實，母相乘得三十六，為法，以列數三乘未并者及法，等數得九，約訖，減四分之三者二，減三分之二者一，并以益三分之一，各平于一十二分之七。

Method: Place '3 parts', '3 parts' and '4 parts' on the right side, and '1 of them', '2 of them' and '3 of them' on the left side. The denominators cross-multiply the numerators. Combine [the products] in an auxiliary position (fu), obtaining 63. Place [this] on the right as the levelling dividend (pingshi). Multiply the denominators together, obtaining 36, as the divisor. With the number of listed [items] (lieshu), 3, multiply those not combined and the divisor; the equal number (dengshu) is found to be 9. When the reduction is completed, subtract 2 from the 3/4, subtract 1 from the 2/3, combine [these] and augment the 1/3 with them; each is level at 7/12.

sunzi-003.7

今有粟一斗。問：為糲米幾何？答曰：六升。

Suppose there is 1 dou (斗) of millet (su). Question: how much coarse grain (limi) does it make? Answer: 6 sheng (升).

sunzi-003.8

術曰：置粟一斗十升，以糲米率三十乘之，得三百升為實，以粟率五十為法，除之，即得。

Method: Place the millet, 1 dou [as] 10 sheng. Multiply it by the rate (lü) for coarse grain, 30, obtaining 300 sheng as the dividend. Take the rate for millet, 50, as the divisor and divide, and [the answer] is obtained.

sunzi-003.9

今有粟二斗一升。問：為粺米幾何？答曰：一斗一升五十分升之一十七。

Suppose there is 2 dou 1 sheng of millet. Question: how much fine grain (baimi) does it make? Answer: 1 dou 1 17/50 sheng.

sunzi-003.10

術曰：置粟數二十一升，以粺米率二十七乘之，得五百六十七升，為實；以粟率五十為法，除之不盡，以法而命分。

Method: Place the amount of millet, 21 sheng. Multiply it by the rate for fine grain, 27, obtaining 567 sheng, as the dividend. Take the rate for millet, 50, as the divisor. Divide; it is not exhausted, so name the parts (mingfen) by the divisor.

sunzi-003.11

今有粟四斗五升。問：為□米幾何？答曰：二斗一升五分升之三。

Suppose there is 4 dou 5 sheng of millet. Question: how much [illegible] grain does it make? Answer: 2 dou 1 3/5 sheng.

sunzi-003.12

術曰：置粟四十五升，以二約□米率二十四，得一十二，乘之，得五百四十升，為實；以二約粟率，五十得二十五，為法，除之，不盡，以等數約之，而命分。

Method: Place the millet, 45 sheng. Reduce by 2 the rate for [illegible] grain, 24, obtaining 12, and multiply by it, obtaining 540 sheng, as the dividend. Reduce the rate for millet by 2: 50 gives 25, as the divisor. Divide; it is not exhausted. Reduce by the equal number, and name the parts.

sunzi-003.13

今有粟七斗九升。問：為御米幾何？答曰：三斗三升一合八勺。

Suppose there is 7 dou 9 sheng of millet. Question: how much imperial grain (yumi) does it make? Answer: 3 dou 3 sheng 1 ge (合) 8 shao (勺).

sunzi-003.14

術曰：置七斗九升，以御米率二十一乘之，得一千六百五十九，為實，以粟率五十除之，即得。

Method: Place 7 dou 9 sheng and multiply it by the rate for imperial grain, 21, obtaining 1659, as the dividend. Divide by the rate for millet, 50, and [the answer] is obtained.

sunzi-003.15

今有屋基，南北三丈，東西六丈，欲以磚瓦砌之，凡積二尺，用磚五枚。問：計幾何？答曰：四千五百枚。

Suppose there is the foundation of a house, 3 zhang (丈) north–south and 6 zhang east–west, which one wishes to lay with bricks and tiles; for every 2 chi (尺) of area, 5 bricks are used. Question: how many in total? Answer: 4500 pieces.

sunzi-003.16

術曰：置東西六丈，以南北三丈乘之，得一千八百尺；以五乘之，得九千尺；以二除之，即得。

Method: Place the east–west 6 zhang and multiply it by the north–south 3 zhang, obtaining 1800 chi. Multiply by 5, obtaining 9000 chi. Divide by 2, and [the answer] is obtained.

sunzi-003.17

今有圓窖，下周二百八十六尺，深三丈六尺。問：受粟幾何？答曰：一十五萬一千四百七十四斛七升二十七分升之一十一。

Suppose there is a round cellar whose lower circumference is 286 chi and whose depth is 3 zhang 6 chi. Question: how much millet does it hold? Answer: 151474 hu (斛) 7 11/27 sheng.

sunzi-003.18

術曰：置周二百八十六尺，自相乘得八萬一千七百九十六尺，以深三丈六尺乘之，得二百九十四萬四千六百五十六；以一十二除之，得二十四萬五千三百八十八尺，以斛法一尺六寸二分除之，即得。

Method: Place the circumference, 286 chi, and multiply it by itself, obtaining 81796 chi. Multiply by the depth, 3 zhang 6 chi, obtaining 2944656. Divide by 12, obtaining 245388 chi. Divide by the hu divisor (hufa), 1 chi 6 cun (寸) 2 fen (分), and [the answer] is obtained.

sunzi-003.19

今有方窖，廣四丈六尺，長五丈四尺，深三丈五尺。問：受粟幾何？答曰：五萬三千六百六十六斛六斗六升三分升之二。

Suppose there is a square cellar, 4 zhang 6 chi wide, 5 zhang 4 chi long and 3 zhang 5 chi deep. Question: how much millet does it hold? Answer: 53666 hu 6 dou 6 2/3 sheng.

sunzi-003.20

術曰：置廣四丈六尺，長五丈四尺，相乘得二千四百八十四尺；以深三丈五尺乘之，得八萬六千九百四十尺，以斛法一尺六寸二分除之，即得。

Method: Place the width, 4 zhang 6 chi, and the length, 5 zhang 4 chi, and multiply them together, obtaining 2484 chi. Multiply by the depth, 3 zhang 5 chi, obtaining 86940 chi. Divide by the hu divisor, 1 chi 6 cun 2 fen, and [the answer] is obtained.

sunzi-003.21

今有圓窖，周五丈四尺，深一丈八尺。問：受粟幾何？答曰：二千七百斛。

Suppose there is a round cellar with a circumference of 5 zhang 4 chi and a depth of 1 zhang 8 chi. Question: how much millet does it hold? Answer: 2700 hu.

sunzi-003.22

術曰：先置周五丈四尺相乘，得二千九百一十六尺，以深一丈八尺乘之，得五萬二千四百八十八尺；以一十二除之，得四千三百七十四尺，以斛法一尺六寸二分除之，即得。

Method: First place the circumference, 5 zhang 4 chi, and multiply [it by itself], obtaining 2916 chi. Multiply by the depth, 1 zhang 8 chi, obtaining 52488 chi. Divide by 12, obtaining 4374 chi. Divide by the hu divisor, 1 chi 6 cun 2 fen, and [the answer] is obtained.

sunzi-003.23

今有圓田，周三百步，徑一百步。問：得田幾何？答曰：三十一畝，奇六十步。

Suppose there is a round field with a circumference of 300 bu (步) and a diameter of 100 bu. Question: how much field is obtained? Answer: 31 mu (畝), with 60 bu over.

sunzi-003.24

術曰：先置周三百步，半之，得一百五十步；又置徑一百步，半之，得五十步，相乘，得七千五百步，以畝法二百四十步除之，即得。

Method: First place the circumference, 300 bu, and halve it, obtaining 150 bu. Further, place the diameter, 100 bu, and halve it, obtaining 50 bu. Multiply them together, obtaining 7500 bu. Divide by the mu divisor (mufa), 240 bu, and [the answer] is obtained.

sunzi-003.25

又術曰：周自相乘，得九萬步，以十二除之，得七千五百步，以畝法除之，得畝數。

Another method: Multiply the circumference by itself, obtaining 90000 bu. Divide by 12, obtaining 7500 bu. Divide by the mu divisor, and the number of mu is obtained.

sunzi-003.26

又術曰：徑自乘，得一萬，以三乘之，得三萬步，四除之，得七千五百步，以畝法除之，得畝數。

Another method: Multiply the diameter by itself, obtaining 10000. Multiply by 3, obtaining 30000 bu. Divide by 4, obtaining 7500 bu. Divide by the mu divisor, and the number of mu is obtained.

sunzi-003.27

今有方田，桑生中央，從角至桑，一百四十七步。問：為田幾何？答曰：一頃八十三畝，奇一百八十步。

Suppose there is a square field with a mulberry tree growing in the centre; from a corner to the mulberry tree is 147 bu. Question: how much field does it make? Answer: 1 qing (頃) 83 mu, with 180 bu over.

sunzi-003.28

術曰：置角至桑一百四十七步，倍之，得二百九十四步，以五乘之，得一千四百七十步，以七除之，得二百一十步，自相乘，得四萬四千一百步，以二百四十步除之，即得。

Method: Place [the distance] from the corner to the mulberry tree, 147 bu, and double it, obtaining 294 bu. Multiply by 5, obtaining 1470 bu. Divide by 7, obtaining 210 bu. Multiply it by itself, obtaining 44100 bu. Divide by 240 bu, and [the answer] is obtained.

sunzi-003.29

今有木，方三尺，高三尺，欲方五寸作枕一枚。問：得幾何？答曰：二百一十六枚。

Suppose there is a piece of wood, 3 chi square and 3 chi high, from which one wishes to make one pillow [for each] 5 cun square. Question: how many are obtained? Answer: 216 pieces.

sunzi-003.30

術曰：置方三尺，自相乘，得九尺，以高三尺乘之，得二十七尺，以一尺木八枕乘之，即得。

Method: Place the square [side], 3 chi, and multiply it by itself, obtaining 9 chi. Multiply by the height, 3 chi, obtaining 27 chi. Multiply by the 8 pillows [from] 1 chi of wood, and [the answer] is obtained.

sunzi-003.31

今有索，長五千七百九十四步，以四除之，得一千四百四十八步，餘二步，以六因之，得一丈二尺，以四除之，得三尺，通計即得。

Suppose there is a rope 5794 bu long. Divide by 4, obtaining 1448 bu, with 2 bu remaining. Multiply (yin) this by 6, obtaining 1 zhang 2 chi. Divide by 4, obtaining 3 chi. Reckoning all together, [the answer] is obtained.

sunzi-003.32

今有堤，下廣五丈，上廣三丈，高二丈，長六十尺，欲以一千尺作一方。問：計幾何？答曰：四十八方。

Suppose there is an embankment, 5 zhang wide at the bottom, 3 zhang wide at the top, 2 zhang high and 60 chi long, and one wishes to take 1000 chi as 1 fang (方). Question: how many in total? Answer: 48 fang.

sunzi-003.33

術曰：置堤，上廣三丈，下廣五丈。并之，得八丈；半之，得四丈。以二丈乘之，得八百尺；以長六十尺乘之，得四萬八千；以一千尺除之（案：原本訛作乘，今改正。），即得。

Method: Place the embankment: the upper width, 3 zhang, and the lower width, 5 zhang. Combine them, obtaining 8 zhang; halve it, obtaining 4 zhang. Multiply by 2 zhang, obtaining 800 chi. Multiply by the length, 60 chi, obtaining 48000. Divide by 1000 chi [Siku editors: the original text wrongly has 'multiply'; it is now corrected.], and [the answer] is obtained.

sunzi-004.1

今有溝，廣十丈，深五丈，長二十丈，欲以千尺作一方。問：得幾何？答曰：一千方。

Suppose there is a ditch, 10 zhang (丈) wide, 5 zhang deep and 20 zhang long, and one wishes to take 1000 chi (尺) to make one fang (方). Question: how many does one get? Answer: 1000 fang.

sunzi-004.2

術曰：置廣一十丈，以深五丈乘之，得五千尺；又以長二十丈乘之，得一百萬尺；以一千除之，即得。

Method: Set out the width, 10 zhang; multiply it by the depth, 5 zhang, obtaining 5000 chi; then multiply it by the length, 20 zhang, obtaining 1,000,000 chi; divide it by 1000, and one obtains it.

sunzi-004.3

今有積，二十三萬四千五百六十七步。問：為方幾何？答曰：四百八十四步九百六十八分步之三百一十一。

Suppose there is an area of 234,567 bu (步). Question: as a square, how much is it? Answer: 484 311/968 bu.

sunzi-004.4

術曰：置積二十三萬四千五百六十七步，為實；次借一算為下法，步之超一位至百而止。上商置四百于實之上（案：上商原本脫上字，今補。），副置四萬于實之下。下法之商，名為方法；命上商四百除實，除訖，倍方法，方法一退（案：原本脫方法二字，今補。），下法再退，復置上商八十以次前商，副置八百于方法之下。下法之上，名為廉法；方廉各命上商八十以除實（案：原本脫實字，今補。），除訖（案：原本脫除字，今補。），倍廉法，從方法，方法一退，下法再退，復置上商四以次前，副置四于方法之下。下法之上，名曰隅法；方廉隅各命上商四以除實，除訖，倍隅法，從方法（案：原本訛此六字，今據術補。），上商得四百八十四，下法得九百六十八，不盡三百一十一，是為方四百八十四步九百六十八分步之三百一十一。

Method: Set out the area, 234,567 bu, as the dividend (shi). Next, borrow one counting rod (suan) as the lower divisor (xiafa); step it forward, skipping one place each time; when it reaches the hundreds, stop. For the upper quotient (shangshang), place 400 above the dividend [Siku editors: in 上商 ('upper quotient') the original text omits the character 上 ('upper'); it is now supplied.], and, as an auxiliary, place 40,000 below the dividend. The quotient of the lower divisor is called the square divisor (fangfa) [Translator's note: 商'quotient' is probably an error for 上'above', as in the parallel sentences.]; apply the upper quotient 400 to divide the dividend. When the division is finished, double the square divisor; move the square divisor back one place [Siku editors: the original text omits the two characters 方法 ('square divisor'); they are now supplied.], and move the lower divisor back two places. Again place the upper quotient 80, following the previous quotient, and, as an auxiliary, place 800 below the square divisor. What is above the lower divisor is called the edge divisor (lianfa); the square and edge divisors each apply the upper quotient 80 to divide the dividend [Siku editors: the original text omits the character 實 ('dividend'); it is now supplied.]. When the division is finished [Siku editors: the original text omits the character 除 ('division'); it is now supplied.], double the edge divisor and join it to the square divisor; move the square divisor back one place, and move the lower divisor back two places. Again place the upper quotient 4, following the previous one, and, as an auxiliary, place 4 below the square divisor. What is above the lower divisor is called the corner divisor (yufa); the square, edge and corner divisors each apply the upper quotient 4 to divide the dividend. When the division is finished, double the corner divisor and join it to the square divisor [Siku editors: in the original text these six characters are corrupt; they are now supplied on the basis of the method.]. The upper quotient gives 484, the lower divisor gives 968, and 311 is not exhausted. This makes the square 484 311/968 bu.

sunzi-004.5

今有積，三萬五千步。問：為圓幾何？答曰：六百四十八步一千二百九十六分步之九十六。（案：六分步原本訛作七分，脫步字，今補正。）

Suppose there is an area of 35,000 bu. Question: as a circle, how much is it? Answer: 648 96/1296 bu. [Siku editors: for 六分步 ('6 parts of a bu') the original text erroneously has 七分 ('7 parts') and omits the character 步 ('bu'); it is now supplied and corrected.]

sunzi-004.6

術曰：置積三萬五千步，以一十二乘之，得四十二萬，為實；次借一算為下法，步之超一位至百而止；上商置六百于實之上，副置六萬于實之下。下法之上，名為方法，命上商六百除實，除訖，倍方法，方法一退，下法再退，復置上商四十以次前商，副置四百于方法之下。下法之上，名為廉法，方廉各命上商四十以除實（案：原本脫四十二字，今補。），除訖，倍廉法，從方法，方法一退，下法再退，復置上商八次前商，副置八于方法之下。下法之上，名為隅法，方廉隅各命上商八以除實，除訖，倍隅法，從方法，上商得六百四十八（案：原本脫得字，今補。），下法得一千二百九十六（案：六原本訛作七，今改正。），不盡九十六，是為方六百四十八步一千二百九十六分步之九十六。（案：九十六分原本訛作九十七分，今改正。）

Method: Set out the area, 35,000 bu; multiply it by 12, obtaining 420,000, as the dividend. Next, borrow one counting rod as the lower divisor; step it forward, skipping one place each time; when it reaches the hundreds, stop. For the upper quotient, place 600 above the dividend, and, as an auxiliary, place 60,000 below the dividend. What is above the lower divisor is called the square divisor; apply the upper quotient 600 to divide the dividend. When the division is finished, double the square divisor; move the square divisor back one place, and move the lower divisor back two places. Again place the upper quotient 40, following the previous quotient, and, as an auxiliary, place 400 below the square divisor. What is above the lower divisor is called the edge divisor; the square and edge divisors each apply the upper quotient 40 to divide the dividend [Siku editors: the original text omits the two characters 四十 ('40'); they are now supplied.]. When the division is finished, double the edge divisor and join it to the square divisor; move the square divisor back one place, and move the lower divisor back two places. Again place the upper quotient 8, following the previous quotient, and, as an auxiliary, place 8 below the square divisor. What is above the lower divisor is called the corner divisor; the square, edge and corner divisors each apply the upper quotient 8 to divide the dividend. When the division is finished, double the corner divisor and join it to the square divisor. The upper quotient gives 648 [Siku editors: the original text omits the character 得 ('gives'); it is now supplied.], the lower divisor gives 1296 [Siku editors: 六 ('6') in the original text is erroneously written as 七 ('7'); it is now corrected.], and 96 is not exhausted. This makes the square 648 96/1296 bu. [Siku editors: 九十六分 ('96 parts') in the original text is erroneously written as 九十七分 ('97 parts'); it is now corrected.]

sunzi-004.7

今有邱田，周六百三十九步，徑三百八十步。問：為田幾何？答曰：二頃五十二畝二百二十五步。

Suppose there is a hill field (qiutian), with circumference 639 bu and diameter 380 bu. Question: how much field does it make? Answer: 2 qing (頃) 52 mu (畝) 225 bu.

sunzi-004.8

術曰：半周，得三百一十九步五分；半徑，得一百九十步；二位相乘，得六萬七百五步；以畝法除之，即得。

Method: Halve the circumference, obtaining 319 bu 5 fen (分); halve the diameter, obtaining 190 bu; multiply the two positions together, obtaining 60,705 bu; divide it by the mu divisor (mufa), and one obtains it.

sunzi-004.9

今有築城，上廣二丈，下廣五丈四尺，高三丈八尺，長五千五百五十尺，秋程人功三百尺。問：須功幾何？答曰：二萬六千一十一功。

Suppose there is a city wall to be built, with upper width 2 zhang, lower width 5 zhang 4 chi, height 3 zhang 8 chi and length 5550 chi; the autumn norm (qiucheng) of a man's labour (rengong) is 300 chi. Question: how much labour (gong) is needed? Answer: 26,011 labours.

sunzi-004.10

術曰：并上下廣，得七十四尺，半之，得三十七尺，以高乘之，得一千四百六尺，又以長乘之，得積七百八十萬三千三百尺，以秋程人功三百尺除之，即得。

Method: Add together the upper and lower widths, obtaining 74 chi; halve it, obtaining 37 chi; multiply it by the height, obtaining 1406 chi; then multiply it by the length, obtaining the volume 7,803,300 chi; divide it by the autumn norm of a man's labour, 300 chi, and one obtains it.

sunzi-004.11

今以穿渠，長二十九里一百四步，上廣一丈二尺六寸，下廣八尺，深一丈八尺，秋程人功三百尺。問：須功幾何？答曰：三萬二千六百四十五功（案：原本訛作三萬二百六十五人，今據術改正。），不盡六十九尺六寸。

Now, for digging a canal, the length is 29 li (里) 104 bu, the upper width 1 zhang 2 chi 6 cun (寸), the lower width 8 chi, the depth 1 zhang 8 chi; the autumn norm of a man's labour is 300 chi. Question: how much labour is needed? Answer: 32,645 labours [Siku editors: the original text erroneously has 三萬二百六十五人 ('30,265 men'); it is now corrected on the basis of the method.], with 69 chi 6 cun not exhausted.

sunzi-004.12

術曰：置里數，以三百步乘之，內零步，六之，得五萬二千八百二十四尺；并上下廣，得二丈六寸，半之，以深乘之，得一百八十五尺四寸；以長乘，得九百七十九萬三千五百六十九尺六寸；以人功三百尺除之，即得。

Method: Set out the number of li; multiply it by 300 bu; put in the odd bu (lingbu); multiply it by 6, obtaining 52,824 chi. Add together the upper and lower widths, obtaining 2 zhang 6 cun; halve it; multiply it by the depth, obtaining 185 chi 4 cun; multiply by the length, obtaining 9,793,569 chi 6 cun; divide it by the man's labour of 300 chi, and one obtains it.

sunzi-004.13

今有錢六千九百三十，欲令二百一十六人作九分分之：八十一人，人與二分；七十二人，人與三分；六十三人，人與四分。問：三種各得幾何？答曰：二分人得錢二十二，三分人得錢三十三，四分人得錢四十四。

Suppose there are 6930 qian (錢) of cash, and one wishes to have 216 people make 9 shares (fen) and divide it: 81 people, each person given 2 shares; 72 people, each person given 3 shares; 63 people, each person given 4 shares. Question: how much does each of the three kinds get? Answer: a 2-share person gets 22 qian, a 3-share person gets 33 qian, a 4-share person gets 44 qian.

sunzi-004.14

術曰：先置八十一人于上，七十二人次之，六十三人在下。頭位以二乘之，得一百六十二；次位以三乘之，得二百一十六；下位以四乘之，得二百五十二。副并三位，得六百三十，為法。又置錢六千九百三十為三位，頭位以一百六十二乘之，得一百一十二萬二千六百六十；又以二百一十六乘中位，得一百四十九萬六千八百八十；又以二百五十二乘下位，得一百七十四萬六千三百六十。各為實，以法六百三十各除之，頭位得一千七百八十二，中位得二千三百七十六，下位得二千七百七十二。各以人數除之，即得。

Method: First place 81 people at the top, 72 people next to it, and 63 people at the bottom. Multiply the head position by 2, obtaining 162; multiply the next position by 3, obtaining 216; multiply the bottom position by 4, obtaining 252. In an auxiliary position add the three positions together, obtaining 630, as the divisor (fa). Again set out the cash, 6930, in three positions; multiply the head position by 162, obtaining 1,122,660; then multiply the middle position by 216, obtaining 1,496,880; then multiply the bottom position by 252, obtaining 1,746,360. Each is taken as a dividend; with the divisor, 630, divide each of them: the head position gives 1782, the middle position gives 2376, the bottom position gives 2772. Divide each by the number of people, and one obtains it.

sunzi-004.15

今有五等諸侯，共分橘子六十顆，人別加三顆。問：五人各得幾何？答曰：公一十八顆，侯一十五顆，伯一十二顆，子九顆，男六顆。

Suppose there are feudal lords of the five ranks who together share 60 oranges, person by person adding 3 oranges. Question: how much does each of the 5 people get? Answer: the duke 18 oranges, the marquis 15 oranges, the earl 12 oranges, the viscount 9 oranges, the baron 6 oranges.

sunzi-004.16

術曰：先置人數別加三顆于下，次六顆，次九顆，次一十二顆，上十五顆；副并之，得四十五，以減六十顆，餘人數除之，得人三顆；各加不并者，上得一十八顆，為公分；次得一十五，為侯分；次得一十二，為伯分；次得九，為子分；下得六，為男分。

Method: First set out, by the number of people, the separately added 3 oranges at the bottom, next 6 oranges, next 9 oranges, next 12 oranges, and at the top 15 oranges. In an auxiliary position add them together, obtaining 45; subtract it from the 60 oranges; divide the remainder by the number of people, obtaining 3 oranges per person. Add it to each of those not added together: the top gets 18 oranges, as the duke's share; the next gets 15, as the marquis's share; the next gets 12, as the earl's share; the next gets 9, as the viscount's share; the bottom gets 6, as the baron's share.

sunzi-005.1

今有甲、乙、丙三人持錢。甲語乙、丙：「各將公等所持錢半以益我，錢成九十。」乙復語甲、丙：「各將公等所持錢，半以益我，錢成七十。」丙復語甲、乙：「各將公等所持錢，半以益我，錢成五十六。」問：三人元持錢各若干？答曰：甲七十二，乙三十二，丙四。

Suppose there are three people, A, B and C, holding cash (qian). A says to B and C: ”Each of you gentlemen, take half of the cash you hold to add to mine, and the cash comes to 90.” B in turn says to A and C: ”Each of you gentlemen, take half of the cash you hold to add to mine, and the cash comes to 70.” C in turn says to A and B: ”Each of you gentlemen, take half of the cash you hold to add to mine, and the cash comes to 56.” Question: How much cash did each of the three people originally hold? Answer: A 72, B 32, C 4.

sunzi-005.2

術曰：先置三人所語為位，以三乘之，各為積：甲得二百七十，乙得二百一十，丙得八十四。又置甲九十，乙七十，丙五十六，各半之。以甲、乙減丙，以甲、丙減乙，以乙、丙減甲，即各得元數。

Method: First set out what the three people said as positions (wei); multiply them by 3, each making a product (ji): A obtains 270, B obtains 210, C obtains 84. [Translator’s note: 3 × 56 is 168; 84 is half of it.] Again set out A 90, B 70, C 56, and halve each. Subtract A and B from C, subtract A and C from B, subtract B and C from A; then the original amount of each is obtained.

sunzi-005.3

今有女子善織，日自倍，五日織通五尺。問：日織幾何？答曰：初日織一寸三十一分寸之一十九，次日織三寸三十一分寸之七，次日織六寸三十一分寸之一十四，次日織一尺二寸三十一分寸之二十八，次日織二尺五寸三十一分寸之二十五。

Suppose there is a woman who is good at weaving, and each day she doubles her own weaving; in 5 days she weaves 5 chi (尺) in all. Question: How much does she weave each day? Answer: On the first day she weaves 1 19/31 cun (寸); on the next day she weaves 3 7/31 cun; on the next day she weaves 6 14/31 cun; on the next day she weaves 1 chi 2 28/31 cun; on the next day she weaves 2 chi 5 25/31 cun.

sunzi-005.4

術曰：各置列衰，副并，得三十一為法。以五尺乘并者，各自為實，實如法而一，即得。

Method: Set out each of the ranked proportions (liecui); combine them in an auxiliary position, obtaining 31 as the divisor (fa). Multiply the ones combined by 5 chi, each separately making a dividend (shi); divide the dividend by the divisor, and it is obtained.

sunzi-005.5

今有人盜庫絹，不知所失幾何。但聞草中分絹，人得六匹，盈六匹；人得七匹，不足七匹。問：人、絹各幾何？答曰：賊一十三人，絹八十四匹。

Suppose there are people who steal silk (juan) from a storehouse, and it is not known how much was lost. One only hears that, dividing the silk in the grass, if each person gets 6 pi (匹), there is an excess of 6 pi; if each person gets 7 pi, there is a deficit of 7 pi. Question: How many are the people and how much is the silk, each? Answer: Thieves, 13 people; silk, 84 pi.

sunzi-005.6

術曰：先置人得六匹于右上，盈六匹于右下；後置人得七匹于左上，不足七匹于左下。維乘之，所得并之為絹；并盈不足為人。

Method: First set out ”each person gets 6 pi” at the upper right, and the excess of 6 pi at the lower right; then set out ”each person gets 7 pi” at the upper left, and the deficit of 7 pi at the lower left. Cross-multiply them (weicheng); combine what is obtained to make the silk; combine the excess and the deficit to make the people.

sunzi-005.7

卷下

Lower Fascicle (juan xia)

sunzi-005.8

今有甲、乙、丙、丁、戊、己、庚、辛、壬、癸九家共輸租：甲出三十五斛，乙出四十六斛，丙出五十七斛，丁出六十八斛，戊出七十九斛，己出八十斛，庚出一百斛，辛出二百一十斛，壬出三百二十五斛。凡九家，共輸租一千斛，僦運直折二百斛外。問：家各幾何？答曰：甲二十八斛，乙三十六斛八斗，丙四十五斛六斗，丁五十四斛四斗，戊六十三斛二斗，己六十四斛，庚八十斛，辛一百六十八斛，壬二百六十斛。

Suppose there are A, B, C, D, E, F, G, H, I, J, nine families, together delivering tax (zu): A puts out 35 hu (斛), B puts out 46 hu, C puts out 57 hu, D puts out 68 hu, E puts out 79 hu, F puts out 80 hu, G puts out 100 hu, H puts out 210 hu, I puts out 325 hu. In all, the nine families together deliver 1000 hu of tax, with 200 hu deducted separately for the cost of hired transport. [Translator's note: ten names are listed, but the text speaks of nine families and gives nine amounts.] Question: How much for each family? Answer: A 28 hu, B 36 hu 8 dou (斗), C 45 hu 6 dou, D 54 hu 4 dou, E 63 hu 2 dou, F 64 hu, G 80 hu, H 168 hu, I 260 hu.

sunzi-005.9

術曰：置甲出三十五斛，以四乘之，得一百四十斛；以五除之，得二十八斛。乙出四十六斛，以四乘之，得一百八十四斛；以五除之，得三十六斛八斗。丙出五十七斛，以四乘之，得二百二十八斛；以五除之，得四十五斛六斗。丁出六十八斛，以四乘之，得二百七十二斛；以五除之，得五十四斛四斗。戊出七十九斛，以四乘之，得三百一十六斛；以五除之，得六十三斛二斗。己出八十斛，以四乘之，得三百二十斛；以五除之，得六十四斛。庚出一百斛，以四乘之，得四百斛；以五除之，得八十斛。辛出二百一十斛，以四乘之，得八百四十斛；以五除之，得一百六十八斛。壬出三百二十五斛，以四乘之，得一千三百斛；以五除之，得二百六十斛。

Method: Set out the 35 hu put out by A; multiply it by 4, obtaining 140 hu; divide it by 5, obtaining 28 hu. B puts out 46 hu: multiply it by 4, obtaining 184 hu; divide it by 5, obtaining 36 hu 8 dou. C puts out 57 hu: multiply it by 4, obtaining 228 hu; divide it by 5, obtaining 45 hu 6 dou. D puts out 68 hu: multiply it by 4, obtaining 272 hu; divide it by 5, obtaining 54 hu 4 dou. E puts out 79 hu: multiply it by 4, obtaining 316 hu; divide it by 5, obtaining 63 hu 2 dou. F puts out 80 hu: multiply it by 4, obtaining 320 hu; divide it by 5, obtaining 64 hu. G puts out 100 hu: multiply it by 4, obtaining 400 hu; divide it by 5, obtaining 80 hu. H puts out 210 hu: multiply it by 4, obtaining 840 hu; divide it by 5, obtaining 168 hu. I puts out 325 hu: multiply it by 4, obtaining 1300 hu; divide it by 5, obtaining 260 hu.

sunzi-005.10

今有丁一千五百萬，出兵四十萬。問：幾丁科一兵？答曰：三十七丁五分。

Suppose there are 15,000,000 adult men (ding), who put out 400,000 soldiers. Question: How many men are assessed for one soldier? Answer: 37 men and 5 tenths (fen).

sunzi-005.11

術曰：置丁一千五百萬為實，以兵四十萬為法。實如法，即得。

Method: Set out the men, 15,000,000, as the dividend (shi); take the soldiers, 400,000, as the divisor (fa). Divide the dividend by the divisor, and it is obtained.

sunzi-005.12

今有平地聚粟，下周三丈六尺，高四尺五寸。問：粟幾何？答曰：一百斛。

Suppose there is millet (su) heaped on level ground, with a lower circumference of 3 zhang (丈) 6 chi and a height of 4 chi 5 cun. Question: How much millet is there? Answer: 100 hu.

sunzi-005.13

術曰：置周三丈六尺，自相乘，得一千二百九十六尺；以高四尺五寸乘之，得五千八百三十二尺；以三十六除之，得一百六十二尺；以斛法一尺六寸二分除之，即得。

Method: Set out the circumference, 3 zhang 6 chi; multiply it by itself, obtaining 1296 chi; multiply this by the height, 4 chi 5 cun, obtaining 5832 chi; divide by 36, obtaining 162 chi; divide by the hu divisor (hufa), 1 chi 6 cun 2 fen (分), and it is obtained.

sunzi-005.14

今有佛書，凡二十九章，章六十三字。問：字幾何？答曰：一千八百二十七。

Suppose there is a Buddhist book with 29 chapters in all, each chapter of 63 characters. Question: How many characters are there? Answer: 1827.

sunzi-005.15

術曰：置二十九章，以六十三字乘之，即得。

Method: Set out the 29 chapters, multiply them by the 63 characters, and it is obtained.

sunzi-005.16

今有棋局，方一十九道。問：用棋幾何？答曰：三百六十一。

Suppose there is a qi board (qiju), 19 lines square. Question: How many pieces are used? Answer: 361.

sunzi-005.17

術曰：置一十九道，自相乘之，即得。

Method: Set out the 19 lines, multiply them by themselves, and it is obtained.

sunzi-005.18

今有租九萬八千七百六十二斛，欲以一車載五十斛。問：用車幾何？答曰：一千九百七十五乘，奇一十二斛。

Suppose there is tax grain (zu) of 98,762 hu, and one wishes to load 50 hu on one cart. Question: How many carts are used? Answer: 1975 carts, with a remainder (ji) of 12 hu.

sunzi-005.19

術曰：置租九萬八千七百六十二斛為實，以一車所載五十斛為法。實如法，即得。

Method: Set out the tax, 98,762 hu, as the dividend (shi); take the 50 hu loaded on one cart as the divisor (fa). Divide the dividend by the divisor, and it is obtained.

sunzi-005.20

今有丁九萬八千七百六十六，凡二十五丁出一兵。問：兵幾何？答曰：三千九百五十人，奇一十六丁。

Suppose there are 98,766 adult men, and in all every 25 men put out one soldier. Question: How many soldiers are there? Answer: 3950 persons, with a remainder of 16 men.

sunzi-005.21

術曰：置丁九萬八千七百六十六為實，以二十五為法。實如法，即得。

Method: Set out the men, 98,766, as the dividend (shi); take 25 as the divisor (fa). Divide the dividend by the divisor, and it is obtained.

sunzi-005.22

今有絹七萬八千七百三十二匹，令一百六十二人分之。問：人得幾何？答曰：四百八十六匹。

Suppose there is silk of 78,732 pi, and 162 people are made to divide it. Question: How much does each person get? Answer: 486 pi.

sunzi-005.23

術曰：置絹七萬八千七百三十二匹為實，以一百六十二人為法。實如法，即得。

Method: Set out the silk, 78,732 pi, as the dividend (shi); take the 162 people as the divisor (fa). Divide the dividend by the divisor, and it is obtained.

sunzi-005.24

今有綿九萬一千一百三十五觔，給與三萬六千四百五十四戶。問：戶得幾何？答曰：二觔八兩。

Suppose there is silk floss (mian) of 91,135 jin (觔), to be given to 36,454 households. Question: How much does each household get? Answer: 2 jin 8 liang (兩).

sunzi-005.25

術曰：置九萬一千一百三十五觔為實，以三萬六千四百五十四戶為法。除之，得二觔，不盡一萬八千二百二十七觔；以一十六乘之，得二十九萬一千六百三十二兩；以戶除之，即得。

Method: Set out 91,135 jin as the dividend (shi); take the 36,454 households as the divisor (fa). Divide, obtaining 2 jin, with 18,227 jin not exhausted (bujin); multiply this by 16, obtaining 291,632 liang; divide it by the households, and it is obtained.

sunzi-005.26

今有粟三千九百九十九斛九斗六升，凡粟九斗易豆一斛。問：計豆幾何？答曰：四千四百四十四斛四斗。

Suppose there is millet of 3999 hu 9 dou 6 sheng (升), and in all 9 dou of millet is exchanged for 1 hu of beans. Question: How much do the beans come to? Answer: 4444 hu 4 dou.

sunzi-005.27

術曰：置粟三千九百九十九斛九斗六升為實，以九斗為法。實如法，即得。

Method: Set out the millet, 3999 hu 9 dou 6 sheng, as the dividend (shi); take 9 dou as the divisor (fa). Divide the dividend by the divisor, and it is obtained.

sunzi-005.28

今有粟二千三百七十四斛，斛加三升。問：共粟幾何？答曰：二千四百四十五斛二斗二升。

Suppose there is millet of 2374 hu, and to each hu 3 sheng are added. Question: How much millet is there altogether? Answer: 2445 hu 2 dou 2 sheng.

sunzi-005.29

術曰：置粟二千三百七十四斛，以一斛三升乘之，即得。

Method: Set out the millet, 2374 hu; multiply it by 1 hu 3 sheng, and it is obtained.

sunzi-006.1

今有粟，三十六萬九千九百八十斛七斗，在倉九年，年斛耗三升。問：一年、九年各耗幾何？答曰：一年耗一萬一千九十九斛四斗二升一合，九年耗九萬九千八百九十四斛七斗八升九合。

Suppose there is millet, 369,980 hu (斛) 7 dou (斗), which lies in a granary for 9 years; each year each hu loses 3 sheng (升). Question: How much is lost in 1 year and in 9 years, respectively? Answer: In 1 year the loss is 11,099 hu 4 dou 2 sheng 1 ge (合); in 9 years the loss is 99,894 hu 7 dou 8 sheng 9 ge.

sunzi-006.2

術曰：置三十六萬九千九百八十斛七斗，以三升乘之，得一年之耗；又以九乘之，即九年之耗。

Method: Set out 369,980 hu 7 dou; multiply it by 3 sheng, obtaining the loss for 1 year; again multiply it by 9: this is the loss for 9 years.

sunzi-006.3

今有貸與人絲五十七觔，限歲出息一十六觔。問：觔息幾何？答曰：四兩五十七分兩之二十八。

Suppose 57 jin (觔) of silk is lent to someone, with a fixed yearly payment of interest of 16 jin. Question: What is the interest per jin? Answer: 4 28/57 liang (兩).

sunzi-006.4

術曰：列限息絲一十六觔，以一十六兩乘之，得二百五十六兩；以貸絲五十七觔除之，不盡，約之，即得。

Method: Lay out the fixed interest silk, 16 jin; multiply it by 16 liang, obtaining 256 liang; divide it by the lent silk, 57 jin; what is not exhausted, reduce it, and you obtain [the answer].

sunzi-006.5

今有三人共車，二車空；二人共車，九人步。問：人與車各幾何？答曰：一十五車，三十九人。

Suppose 3 people share a cart, and 2 carts are empty; 2 people share a cart, and 9 people walk. Question: How many people and carts are there, respectively? Answer: 15 carts, 39 people.

sunzi-006.6

術曰：置二人，以三乘之，得六，加步者九人，得車一十五。欲知人者，以二乘車，加九人，即得。

Method: Set out 2 people; multiply by 3, obtaining 6; add the 9 people who walk, obtaining 15 carts. If you wish to know the people, multiply the carts by 2 and add 9 people, and you obtain [the answer].

sunzi-006.7

今有粟一十二萬八千九百四十斛九斗三合，出與人買絹，一匹直粟三斛五斗七升。問：絹幾何？答曰：三萬六千一百一十七匹三丈六尺。

Suppose there is millet, 128,940 hu 9 dou 3 ge, given out to people to buy silk; 1 pi (匹) is worth 3 hu 5 dou 7 sheng of millet. Question: How much silk? Answer: 36,117 pi 3 zhang (丈) 6 chi (尺).

sunzi-006.8

術曰：置粟一十二萬八千九百四十斛九斗三合為實，以三斛五斗七升為法，除之，得匹；餘四十之，所得又以法除之，即得。

Method: Set out the millet, 128,940 hu 9 dou 3 ge, as the dividend (shi); take 3 hu 5 dou 7 sheng as the divisor (fa); divide, obtaining the pi; multiply the remainder by 40, and divide what is obtained again by the divisor, and you obtain [the answer].

sunzi-006.9

今有婦人河上蕩梧，津吏問曰：「梧何以多?」婦人曰：「家有客。」津吏曰：「客幾何?」婦人曰：「二人共飯，三人共羹，四人共肉，凡用梧六十五，不知客幾何?」答曰：「六十人。」

Suppose a woman is rinsing cups at the river. The ferry official asks: ”Why are the cups so many?” The woman says: ”There are guests at home.” The ferry official says: ”How many guests?” The woman says: ”2 people share [a cup of] rice, 3 people share [a cup of] broth, 4 people share [a cup of] meat; altogether 65 cups are used. I do not know: how many guests are there?” Answer: ”60 people.”

sunzi-006.10

術曰：置六十五梧，以十二乘之，得七百八十；以十三除之，即得。

Method: Set out the 65 cups; multiply by 12, obtaining 780; divide by 13, and you obtain [the answer].

sunzi-006.11

今有木，不知長短。引繩度之，餘繩四尺五寸；屈繩量之，不足一尺。問：幾何？答曰：六尺五寸。

Suppose there is a piece of wood whose length is not known. Stretching a rope to measure it, 4 chi 5 cun (寸) of rope is left over; folding the rope to measure it, it falls short by 1 chi. Question: How much [is it]? Answer: 6 chi 5 cun.

sunzi-006.12

術曰：置餘繩四尺五寸，加不足一尺，共五尺五寸；倍之，得一丈一尺；減四尺五寸，即得。

Method: Set out the leftover rope, 4 chi 5 cun, and add the shortfall, 1 chi: together 5 chi 5 cun; double it, obtaining 1 zhang 1 chi; subtract 4 chi 5 cun, and you obtain [the answer].

sunzi-006.13

今有器中米，不知其數。前人取半，中人三分取一，後人四分取一，餘米一斗五升。問：米幾何？答曰：六斗。

Suppose there is rice in a vessel, its quantity not known. The first person takes half; the middle person takes 1 of 3 parts; the last person takes 1 of 4 parts; 1 dou 5 sheng of rice remains. Question: How much rice? Answer: 6 dou.

sunzi-006.14

術曰：置餘米一斗五升，以六乘之，得九斗；以二除之，得四斗五升；以四乘之，得一斛八斗；以三除之，即得。

Method: Set out the remaining rice, 1 dou 5 sheng; multiply by 6, obtaining 9 dou; divide by 2, obtaining 4 dou 5 sheng; multiply by 4, obtaining 1 hu 8 dou; divide by 3, and you obtain [the answer].

sunzi-006.15

今有黃金一觔，直錢一十萬。問：兩直幾何？答曰：六千二百五十錢。

Suppose 1 jin of yellow gold is worth 100,000 qian (錢). Question: What is a liang worth? Answer: 6,250 qian.

sunzi-006.16

術曰：置錢一十萬，以一十六兩除之，即得。

Method: Set out the 100,000 qian; divide by 16 liang, and you obtain [the answer].

sunzi-006.17

今有錦一匹，直錢一萬八千。問：丈、尺、寸各直幾何？答曰：丈四千五百錢，尺四百五十錢，寸四十五錢。

Suppose there is 1 pi of brocade, worth 18,000 qian. Question: What are a zhang, a chi and a cun each worth? Answer: a zhang, 4,500 qian; a chi, 450 qian; a cun, 45 qian.

sunzi-006.18

術曰：置錢一萬八千，以四除之，得一丈之直；一退、再退，得尺、寸之直。

Method: Set out the 18,000 qian; divide by 4, obtaining the value of 1 zhang; move it back (tui) once and move it back a second time, obtaining the values of a chi and a cun.

sunzi-006.19

今有地，長一千步，廣五百步，尺有鶉，寸有鷃。問：鶉、鷃各幾何？答曰：鶉一千八百萬，鷃一億八千萬。

Suppose there is a piece of land, 1,000 bu (步) long and 500 bu wide; in each chi there is a quail (chun), in each cun there is a yan bird (鷃). Question: How many quails and yan birds are there, respectively? Answer: quails, 18,000,000; yan birds, 180,000,000.

sunzi-006.20

術曰：置長一千步，以廣五百步乘之，得五十萬；以三十六乘之，得一千八百萬尺，即得鶉數；上十之，即得鷃數。

Method: Set out the length, 1,000 bu; multiply by the width, 500 bu, obtaining 500,000; multiply by 36, obtaining 18,000,000 chi: this gives the number of quails; raise it tenfold, and you obtain the number of yan birds.

sunzi-006.21

今有六萬口：上口三萬人，日食九升；中口二萬人，日食七升；下口一萬人，日食五升。問：上、中、下口共食幾何？答曰：四千六百斛。

Suppose there are 60,000 mouths: upper mouths, 30,000 people, eating 9 sheng a day; middle mouths, 20,000 people, eating 7 sheng a day; lower mouths, 10,000 people, eating 5 sheng a day. Question: How much do the upper, middle and lower mouths eat altogether? Answer: 4,600 hu.

sunzi-006.22

術曰：各置口數，以日食之數乘之，所得并之，即得。

Method: Set out each number of mouths; multiply by the amount eaten per day; add together what is obtained, and you obtain [the answer].

sunzi-006.23

今有方物一束，外周一市有三十二枚。問：積幾何？答曰：八十一枚。

Suppose there is a bundle of square objects; the outer perimeter, one circuit, has 32 pieces. [Translator's note: 市"market" here evidently stands for 市"circuit".] Question: What is the area? Answer: 81 pieces.

sunzi-006.24

術曰：重置二位，左位減八，餘加右位，至盡虛，加一，即得。

Method: Set it out twice, in two positions; from the left position subtract 8, and add the remainder to the right position, until it is exhausted and empty; add 1, and you obtain [the answer].

sunzi-006.25

今有竿，不知長短。度其影，得一丈五尺；別立一表，長一尺五寸，影得五寸。問：竿長幾何？答曰：四丈五尺。

Suppose there is a pole whose length is not known. Measuring its shadow, one obtains 1 zhang 5 chi; separately one sets up a gnomon (biao), 1 chi 5 cun long, whose shadow comes to 5 cun. Question: How long is the pole? Answer: 4 zhang 5 chi.

sunzi-006.26

術曰：置竿影一丈五尺，以表長一尺五寸乘之，上十之，得二十二丈五尺；以表影五寸除之，即得。

Method: Set out the pole's shadow, 1 zhang 5 chi; multiply by the gnomon's length, 1 chi 5 cun; raise it tenfold, obtaining 22 zhang 5 chi; divide by the gnomon's shadow, 5 cun, and you obtain [the answer].

sunzi-006.27

今有物，不知其數。三三數之，賸二；五五數之，賸三；七七數之，賸二。問：物幾何？答曰：二十三。

Suppose there are things whose number is not known. Counting them by threes, 2 are left over; counting them by fives, 3 are left over; counting them by sevens, 2 are left over. Question: How many things are there? Answer: 23.

sunzi-006.28

術曰：三三數之，賸二，置一百四十；五五數之，賸三，置六十三；七七數之，賸二，置三十。并之，得二百三十三，以二百一十減之，即得。凡三三數之，賸一，則置七十；五五數之，賸一，則置二十一；七七數之，賸一，則置十五。一百六以上，以一百五減之，即得。

Method: Counting by threes, 2 are left over: set out 140; counting by fives, 3 are left over: set out 63; counting by sevens, 2 are left over: set out 30. Add them, obtaining 233; subtract 210 from it, and you obtain [the answer]. In general, when counting by threes 1 is left over, set out 70; when counting by fives 1 is left over, set out 21; when counting by sevens 1 is left over, set out 15. If [the sum] is 106 or more, subtract 105 from it, and you obtain [the answer].

sunzi-006.29

今有獸，六首四足；禽，二首二足。上有七十六首，下有四十六足。問：禽、獸各幾何？答曰：八獸、七禽。

Suppose there are beasts with 6 heads and 4 feet, and birds with 2 heads and 2 feet. Above there are 76 heads; below there are 46 feet. Question: How many birds and beasts are there, respectively? Answer: 8 beasts, 7 birds. [Translator's note: with 2 heads per bird, the answer does not give 76 heads; the figures as given are inconsistent.]

sunzi-006.30

術曰：倍足以減首，餘半之，即獸；以四乘獸，減足，餘半之，即禽。

Method: Double the feet and subtract the heads; halve the remainder: this is the beasts. Multiply the beasts by 4 and subtract from the feet; halve the remainder: this is the birds.

sunzi-006.31

今有甲、乙二人持錢，各不知數。甲得乙中半，可滿四十八；乙得甲大半，亦滿四十八。問：甲、乙二人元持錢各幾何？答曰：甲持錢三十六，乙持錢二十四。

Suppose two people, Jia (甲) and Yi (乙), hold cash, the amount of each unknown. If Jia obtains the middle half (zhongban) of Yi's, he can make up a full 48; if Yi obtains the greater half (daban) of Jia's, he also makes up a full 48. [Translator's note: zhongban is 1/2; daban is 2/3.] Question: How much cash did Jia and Yi originally hold, respectively? Answer: Jia held 36 qian; Yi held 24 qian.

sunzi-006.32

術曰：如方程求之。置二甲、一乙、錢九十六于右方，置二甲、三乙、錢一百四十四于左方。以右方二乘左方，上得四，中得六，下得二百八十八錢；以左方二乘右方，上得四，中得二，下得九十六（案：近刻脫此十八字，今據術補。）；以右行再減左行，左上空，中餘四，以為法；下餘九十六錢，為實。上法下實，得二十四錢，為乙錢。以減右下九十六，餘七十二為實；以右上二甲為法，上法下實，得三十六，為甲錢也。

Method: Seek it as with rectangular arrays (fangcheng). Set out 2 Jia, 1 Yi and 96 qian on the right; set out 2 Jia, 3 Yi and 144 qian on the left. Multiply the left by the 2 on the right: at the top one obtains 4, in the middle 6, at the bottom 288 qian. Multiply the right by the 2 on the left: at the top one obtains 4, in the middle 2, at the bottom 96 [Siku editors: Recent printings omit these 18 characters; they are now supplied on the basis of the method.] [Translator's note: these supplied figures do not fit the working that follows; twice 96 would be 192.]; subtract the right column twice from the left column: the upper left is empty, in the middle 4 remains, which is taken as the divisor;

at the bottom 96 qian remains, as the dividend. With the divisor above and the dividend below, one obtains 24 qian, which is Yi's cash. Subtract it from the 96 at the lower right, leaving 72 as the dividend; take the 2 Jia at the upper right as the divisor; with the divisor above and the dividend below, one obtains 36, which is Jia's cash.

sunzi-006.33

今有百鹿入城，家取一鹿，不盡；又三家共一鹿，適盡。問：城中家幾何？答曰：七十五家。

Suppose 100 deer enter a city. Each household takes 1 deer, and they are not exhausted; further, every 3 households share 1 deer, and they are exactly exhausted. Question: How many households are there in the city? Answer: 75 households.

sunzi-006.34

術曰：以盈不足取之。假令七十二家，鹿不盡四；令之九十家，鹿不足二十。置七十二于右上，盈四于右下；置九十于左上，不足二十于左下。維乘之，所得并為實；並盈不足為法，除之，即得。

Method: Obtain it by excess and deficit (ying buzu). Suppose 72 households: 4 deer are not exhausted; let it be 90 households: the deer fall short by 20. Set 72 at the upper right and the excess 4 at the lower right; set 90 at the upper left and the deficit 20 at the lower left. Cross-multiply them (weicheng); add together what is obtained as the dividend; add together the excess and the deficit as the divisor; divide, and you obtain [the answer].

sunzi-007.1

今有三雞共啄粟一千一粒，雛啄一，母啄二，翁啄四。主責本粟，三雞主各償幾何？答曰：雞雛主一百四十三，雞母主二百八十六，雞翁主五百七十二。

Suppose there are 3 chickens that together peck 1001 grains of millet: the chick pecks 1, the hen pecks 2, the cock pecks 4. The owner demands [the return of] the original millet. How much must the owners of the 3 chickens each repay? Answer: the owner of the chick, 143; the owner of the hen, 286; the owner of the cock, 572.

sunzi-007.2

術曰：置粟一千一粒為實，副并三雞所啄粟七粒為法，除之，得一百四十三粒為雞雛主所償之數，遞倍之，即得母、翁主所償之數。

Method: Set up the millet, 1001 grains, as the dividend (shi). In an auxiliary position (fu), add together the millet pecked by the 3 chickens, 7 grains, as the divisor (fa). Divide, obtaining 143 grains as the amount to be repaid by the owner of the chick. Double it successively, and you obtain the amounts to be repaid by the owners of the hen and the cock.

sunzi-007.3

今有雉、兔同籠，上有三十五頭，下有九十四足。問：雉、兔各幾何？答曰：雉二十三，兔一十二。

Suppose there are pheasants and rabbits in the same cage: above there are 35 heads, below there are 94 feet. Question: How many pheasants and how many rabbits are there? Answer: pheasants, 23; rabbits, 12.

sunzi-007.4

術曰：上置三十五頭，下置九十四足。半其足，得四十七，以少減多，再命之：上三除下三，上五除下五；下有一除上一，下有二除上二，即得。

Method: Above, set up the 35 heads; below, set up the 94 feet. Halve the feet, obtaining 47. Subtract the lesser from the greater, applying it twice (zai ming zhi): the 3 above removes (chu) 3 below, the 5 above removes 5 below; there being 1 below, it removes 1 above; there being 2 below, it removes 2 above. Then you obtain [the answer].

sunzi-007.5

又術曰：上置頭，下置足，半其足，以頭除足，以足除頭，即得。

Another method: Above, set up the heads; below, set up the feet. Halve the feet; remove the heads from the feet; remove the feet from the heads; and you obtain [the answer].

sunzi-007.6

今有九里渠，三寸魚頭，頭相次。問：魚得幾何？答曰：五萬四千。

Suppose there is a channel of 9 li (里), [with] 3-cun (寸) fish heads, the heads following one another. Question: How many fish are obtained? Answer: 54,000.

sunzi-007.7

術曰：置九里，以三百步乘之，得二千七百步；又以六尺乘之，得一萬六千二百尺；上十之，得一十六萬二千寸；以魚三寸除之，即得。

Method: Set up the 9 li and multiply by 300 bu (步), obtaining 2700 bu. Again multiply by 6 chi (尺), obtaining 16,200 chi. Raise it tenfold (shang shi zhi), obtaining 162,000 cun. Divide by the fish's 3 cun, and you obtain [the answer].

sunzi-007.8

今有長安、洛陽相去九百里，車輪一匝一丈八尺。欲自洛陽至長安。問：輪匝幾何？答曰：九萬匝。

Suppose there are Chang'an and Luoyang, 900 li apart from each other, and one revolution (za) of a cart wheel is 1 zhang (丈) 8 chi. One wishes [to go] from Luoyang to Chang'an. Question: How many revolutions does the wheel make? Answer: 90,000 revolutions.

sunzi-007.9

術曰：置九百里，以三百步乘之，得二十七萬步；又以六尺乘之，得一百六十二萬尺；以車輪一丈八尺為法，除之，即得。

Method: Set up the 900 li and multiply by 300 bu, obtaining 270,000 bu. Again multiply by 6 chi, obtaining 1,620,000 chi. Take the cart wheel's 1 zhang 8 chi as the divisor (fa) and divide, and you obtain [the answer].

sunzi-007.10

今有出門望見九堤，堤有九木，木有九枝，枝有九巢，巢有九禽，禽有九雛，雛有九毛，毛有九色。問：各幾何？答曰：木八十一，枝七百二十九，巢六千五百六十一，禽五萬九千四十九，雛五十三萬一千四百四十一，毛四百七十八萬二千九百六十九，色四千三百四萬六千七百二十一。

Suppose there is [someone who] goes out of the gate and sees 9 embankments in the distance. Each embankment has 9 trees, each tree has 9 branches, each branch has 9 nests, each nest has 9 birds, each bird has 9 chicks, each chick has 9 feathers, each feather has 9 colours. Question: How many of each? Answer: trees, 81; branches, 729; nests, 6561; birds, 59,049; chicks, 531,441; feathers, 4,782,969; colours, 43,046,721.

sunzi-007.11

術曰：置九堤，以九乘之，得木之數；又以九乘之，得枝之數；又以九乘之，得巢之數；又以九乘之，得禽之數；又以九乘之，得雛之數；又以九乘之，得毛之數；又以九乘之，得色之數。

Method: Set up the 9 embankments and multiply by 9, obtaining the number of trees; again multiply by 9, obtaining the number of branches; again multiply by 9, obtaining the number of nests; again multiply by 9, obtaining the number of birds; again multiply by 9, obtaining the number of chicks; again multiply by 9, obtaining the number of feathers; again multiply by 9, obtaining the number of colours.

sunzi-007.12

今有三女，長女五日一歸，中女四日一歸，少女三日一歸。問：三女幾何日相會？答曰：六十日。

Suppose there are 3 daughters: the eldest daughter returns once every 5 days, the middle daughter returns once every 4 days, the youngest daughter returns once every 3 days. Question: After how many days do the 3 daughters meet? Answer: 60 days.

sunzi-007.13

術曰：置長女五日、中女四日、少女三日于右方，各列一算于左方，維乘之，各得所到數：長女十二到，中女十五到，少女二十到。又各以歸日乘到數，即得。

Method: Set up the eldest daughter's 5 days, the middle daughter's 4 days and the youngest daughter's 3 days on the right. On the left lay out 1 counting rod (suan) for each. Cross-multiply them (wei cheng), and you obtain for each the number of arrivals: the eldest daughter arrives 12 times, the middle daughter arrives 15 times, the youngest daughter arrives 20 times. Again, for each, multiply the number of arrivals by the days of return, and you obtain [the answer].

sunzi-007.14

今有孕婦，行年二十九歲，難九月，未知所生？答曰：生男。

Suppose there is a pregnant woman whose current age is 29 years. Her difficulty (nan) is [in] the 9th month, and it is not known what she will give birth to. [Translator's note: 難 nan, "difficulty", here means childbirth.] Answer: She will give birth to a boy.

sunzi-007.15

術曰：置四十九，加難月，減行年，所餘以天除一，地除二，人除三，四時除四，五行除五，六律除六，七星除七，八風除八，九州除九。其不盡者，奇則為男，耦則為女。

Method: Set up 49, add the month of difficulty, and subtract the current age. From what remains, remove 1 for Heaven, remove 2 for Earth, remove 3 for Man, remove 4 for the Four Seasons, remove 5 for the Five Phases, remove 6 for the Six Pitch-pipes, remove 7 for the Seven Stars, remove 8 for the Eight Winds, remove 9 for the Nine Provinces. As for what is not exhausted: if it is odd, it is a boy; if it is even, it is a girl.

Sources

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